District-Level Disparities in Household Consumption: A Bayesian Small Area Estimation Analysis

Official Statistics Project

Pratyush Patra (BS2336)

July 30, 2026

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References

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1 Research Objective

Household Consumption Expenditure is the primary proxy for economic well-being in India. While the NSSO provides robust estimates of Monthly Per Capita Consumption Expenditure (MPCE) at the state level, district-level estimates—especially when disaggregated by sector—are often unreliable due to small sample sizes. This study aims to: **Overcome Small Sample Limitations** using **Small Area Estimation** (SAE) with Census 2011 auxiliary data; **Quantify the Structural Gap** using a Bayesian Hierarchical Model; and **Map Regional Heterogeneity to identify inequality hotspots**.

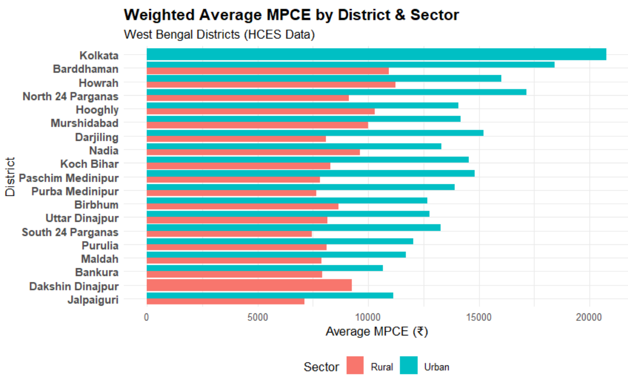
2 Literature Review

The methodological choice to employ a Bayesian Small Area Estimation framework for this study is strongly motivated by recent empirical successes in the Indian context, particularly regarding the utilization of National Sample Survey Office (NSSO) data. The feasibility of generating reliable district-level estimates by linking NSSO Household Consumption Expenditure Survey (HCES) data with Census auxiliary variables was robustly demonstrated by Guha and Chandra [2021] and Chandra and Verma [2022]. Their work on mapping food insecurity in Uttar Pradesh highlighted that while direct survey estimates are often unstable due to small sample sizes, model-based integration of census covariates significantly reduces standard errors. Furthermore, the shift from frequentist prediction to Bayesian inference is supported by Anjoy et al. [2019], who applied a Hierarchical Bayes (HB) model to estimate poverty incidence in Odisha. They argued that the Bayesian framework offers more coherent uncertainty quantification for complex socio-economic indicators compared to traditional frequentist approaches.

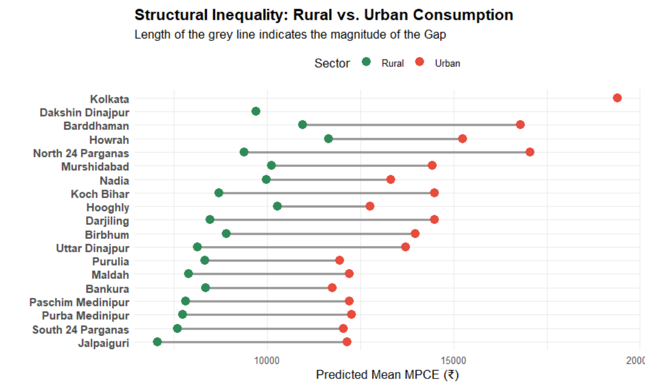
3 Results and Discussions

This study employed a Unit-Level Bayesian Hierarchical Model with random slopes, to address the instability of direct survey estimates in districts with limited sample sizes. This framework integrates household-level demographic characteristics with district-level Census auxiliary variables, mathematically "borrowing strength" across areas to stabilize predictions. Crucially, the inclusion of random slopes allows the urban-rural consumption gap to vary spatially, capturing local economic heterogeneities rather than imposing a uniform state-wide trend. The model demonstrated excellent convergence, with Potential Scale Reduction Factors ($\hat{R}$) for all parameters stabilizing at ≤ 1.01 and effective sample sizes (n_{eff}) exceeding 400. Before moving on to the analysis we take a look at the emperical values of MPCE across districts of West Bengal(figure 1a).

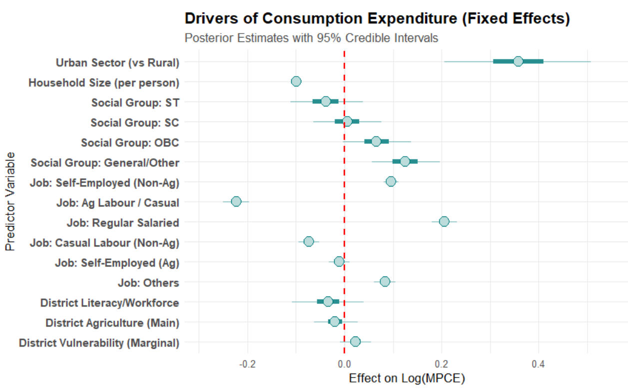
Controlling for district characteristics, the model's fixed effects reveal significant structural determinants of consumption expenditure (Table 3 and figure 1c). After controlling for district-level heterogeneity, the "Urban Pre-



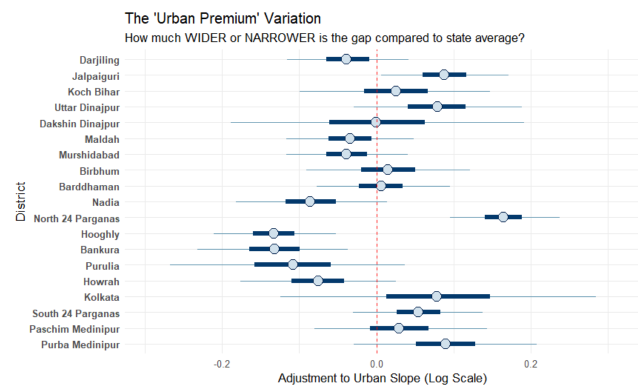
(a) *Weighted Average MPCE*



(b) *Structural Inequality (Rural vs. Urban)*



(c) *Fixed Effects (Drivers of Consumption)*



(d) *Variation in Urban Premium*

Figure 1: *Analysis of Consumption Expenditure and Inequality in West Bengal*

mium" is substantial and statistically significant: an average urban household consumes approximately 43.1% more than a comparable rural household. See figure 1d for district-wise effect. However, occupation appears to be a more profound driver of inequality than geography alone. Households engaged in casual labor face a severe consumption penalty, spending 20.0% less than the reference group, whereas those with regular salaried jobs enjoy a 22.8% premium. This creates a stark consumption divide of over 40% based purely on employment stability. Additionally, household size has a significant negative effect (-9.4%), reflecting economies of scale, while minor social groups like Scheduled Tribe (ST) households show a consumption deficit of 3.7%, though this effect is statistically insignificant. Effect of the auxiliary variables are also insignificant.

A key contribution of this study is quantifying the variability of the rural-urban gap across districts (Table 1 and figure 1b). The Random Slope model reveals that this divide is not uniform.Industrial and peri-urban districts exhibit the widest gaps. North 24 Parganas stands out with an urban premium of 81.7%, followed closely by Jalpaiguri (72.0%) and Darjiling (71.3%). This suggests that in these regions, urban economic growth—likely driven by industrial or tourism sectors—has disproportionately benefited urban residents, leaving rural areas behind.In contrast, districts like Murshidabad (38.8%) and Purulia (43.3%) show a relatively narrower gap. While this indicates lower inequality, it often reflects a scenario where urban centers are not significantly wealthier than their rural hinterlands, pointing to a need for broad-based regional development rather than just rural-focused interventions.

Beyond demographic composition, the model's random intercepts identify districts that are structurally wealthier or poorer than the state average (Table 2). Howrah (+20.4%) and Purulia (+20.1%) show surprisingly high structural wealth effects, meaning households in these districts consume significantly more than their demographic profiles would predict. Conversely, North 24 Parganas (−17.5%) and Jalpaiguri (−21.9%) underperform relative to their potential, suggesting that high intra-district inequality or unmeasured costs of living may be eroding the consumption power of average households in these technically "wealthy" districts.

Table 1: *District Inequality Ranking*

District	Rural (Rs.)	Urban (Rs.)	Prem. (%)	Gap Category
North 24 Parganas	9,383	17,052	81.7	High Inequality
Jalpaiguri	7,064	12,153	72.0	High Inequality
Darjiling	8,462	14,491	71.3	High Inequality
Uttar Dinajpur	8,139	13,705	68.4	High Inequality
Koch Bihar	8,704	14,486	66.4	High Inequality
South 24 Parganas	7,600	12,045	58.5	High Inequality
Purba Medinipur	7,744	12,268	58.4	High Inequality
Birbhum	8,912	13,982	56.9	High Inequality
Paschim Medinipur	7,819	12,200	56.0	High Inequality
Maldah	7,896	12,207	54.6	High Inequality
Barddhaman	10,948	16,789	53.3	High Inequality
Purulia	8,337	11,944	43.3	High Inequality
Murshidabad	10,111	14,437	42.8	High Inequality
Bankura	8,346	11,738	40.6	High Inequality
Nadia	9,971	13,313	33.5	Moderate Inequality
Howrah	11,657	15,245	30.8	Moderate Inequality
Hooghly	10,286	12,758	24.0	Moderate Inequality
Dakshin Dinajpur	9,696	NA		N/A (Rural Only)
Kolkata	NA	19,400		N/A (Urban Only)

Table 2: *Structural Wealth*

District	Wealth (%)	Status
Howrah	20.4	Above Average
Purulia	20.1	Above Average
Murshidabad	19.6	Above Average
Hooghly	19.5	Above Average
Barddhaman	16.2	Above Average
Bankura	11.7	Above Average
Nadia	11.5	Above Average
Darjiling	11.2	Above Average
Kolkata	6.7	Above Average
Dakshin Dinajpur	-0.7	Below Average
Birbhum	-1.8	Below Average
Maldah	-2.7	Below Average
Koch Bihar	-4.4	Below Average
Uttar Dinajpur	-7.9	Below Average
Paschim Medinipur	-12.9	Below Average
North 24 Parganas	-17.5	Below Average
South 24 Parganas	-20.9	Below Average
Jalpaiguri	-21.9	Below Average
Purba Medinipur	-22.1	Below Average

Table 3: *Regression Results (Drivers of Consumption)*

Predictor	Interpretation	Sig.
Living in Urban Area	Increases consumption by 43.1% (22.8% to 66.2%)	Yes
Each Additional Family Member	Decreases consumption by 9.4% (-9.7% to -9.1%)	Yes
Social Group: ST (Scheduled Tribe)	Decreases consumption by 3.7% (-10.6% to 3.8%)	No
Social Group: SC (Scheduled Caste)	Increases consumption by 0.6% (-6.2% to 8%)	No
Social Group: OBC	Increases consumption by 6.8% (-0.3% to 14.7%)	No
Social Group: General/Other	Increases consumption by 13.3% (5.8% to 21.7%)	Yes
Job: Self-Employed (Non-Ag)	Increases consumption by 10.1% (8.5% to 11.8%)	Yes
Job: Casual Labour	Decreases consumption by 20% (-22.1% to -17.8%)	Yes
Job: Regular Salaried	Increases consumption by 22.8% (19.8% to 26%)	Yes
Job: Casual Labour (Non-Ag)	Decreases consumption by 7% (-9% to -5.1%)	Yes
Job: Self-Employed (Ag)	Decreases consumption by 1.1% (-3.2% to 1.1%)	No
Job: Others	Increases consumption by 8.7% (6.2% to 11.1%)	Yes
District Literacy/Workforce Score	Decreases consumption by 3.2% (-10.2% to 4.1%)	No
District Agricultural Score	Decreases consumption by 1.9% (-6.1% to 2.8%)	No
District Vulnerability Score	Increases consumption by 2.4% (-0.9% to 5.6%)	No

This study confirms a significant structural urban consumption advantage in West Bengal (43%), but highlights that occupation (Casual Labour vs. Salaried) is an equally critical driver. The use of SAE reveals spatial heterogeneity: industrial districts show higher divergence than agricultural ones. Policy must target occupational stability in high-gap districts.